

The empirical recurrence for OEIS A258534, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A258534 records “Empirical recurrence of order 61” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 605$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 17$, so the recurrence is proved for every $n \geq 78$. Its last coefficient is $c_{61} = 4947802324992 \neq 0$, so no further tail satisfies anything shorter; any order-61 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A258534 is “Number of $(5+2) \times (n+2)$ 0..1 arrays with every 3×3 subblock sum of the two maximums of the central row and central column plus the two minimums of the diagonal and antidiagonal nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

294960, 1323419, 3806582, 16029944, 65947572, 288227428, 1319588040, 6258499571, ...

The entry states:

Empirical recurrence of order 61 (see link above).

As of the “Last modified” line on the live entry (Apr 18 2023 15:54:53, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 605$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 605$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 61: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 17$ is the first whose minimal order is exactly the stated 61.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1210$ exact terms from $n = 17$ on, it returns order 61 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{61} c_i a(n-i),$$

c_1	8	c_{17}	333455264	c_{33}	−16512679281776	c_{49}	200482555428864
c_2	10	c_{18}	33154890	c_{34}	−28727297989632	c_{50}	−146895229943808
c_3	−161	c_{19}	−2388696589	c_{35}	−18529567470336	c_{51}	−231265953382400
c_4	−57	c_{20}	2369112291	c_{36}	68336824134400	c_{52}	96145309171712
c_5	746	c_{21}	17735972322	c_{37}	87444083931648	c_{53}	63887839854592
c_6	5556	c_{22}	10423157964	c_{38}	−57791850261504	c_{54}	−8884542504960
c_7	−17025	c_{23}	−126471060821	c_{39}	−116999178443776	c_{55}	61832765112320
c_8	−5434	c_{24}	−144255572088	c_{40}	−36441592848384	c_{56}	−33672543600640
c_9	87536	c_{25}	362724798184	c_{41}	−5362457112576	c_{57}	−78346645929984
c_{10}	−52196	c_{26}	987886341184	c_{42}	156136817000448	c_{58}	47004122087424
c_{11}	−1008074	c_{27}	−159092906984	c_{43}	214573459668992	c_{59}	25563645345792
c_{12}	−1133842	c_{28}	−3516939690552	c_{44}	−173792269565952	c_{60}	−24739011624960
c_{13}	15595944	c_{29}	−2225166457988	c_{45}	−254696142667776	c_{61}	4947802324992
c_{14}	1502800	c_{30}	4243653613440	c_{46}	33915983888384		
c_{15}	−52424570	c_{31}	11418888231600	c_{47}	53693947052032		
c_{16}	−106903181	c_{32}	3670670884832	c_{48}	104151547641856		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 78$, and it is the recurrence OEIS A258534 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 17$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{61} - c_1x^{61-1} - \dots - c_{61}$. Its constant term is $-c_{61} = -4947802324992$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 61 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 61, so $\deg q = 61$, and it is monic. It holds from some index t on. From $\max(t, 17)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 61, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 61, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4947802324992.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-61 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A258534.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.